Electromagnetic Waves

The following subjects will be covered

- EM Wave Propagation in lossy Media (1D).

The Plane-wave Solution in Lossy Media

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}$$

Consider a source-free $\rho_v = 0$
and lossy (conducting) $\sigma \neq 0$ medium

Time-harmonic waves are generated by time-harmonic sources.

$$\vec{E} = \vec{E}_s\, e^{j\omega t}$$

$$\vec{\nabla}^2\vec{E}_s \cdot e^{j\omega t} = \mu\sigma\vec{E}_s\,(j\omega)\cdot e^{j\omega t} + \mu\varepsilon\vec{E}_s\,(j\omega)^2\cdot e^{j\omega t}$$

$$\vec{\nabla}^2\vec{E}_s = j\omega\mu\sigma\vec{E}_s + \mu\varepsilon\,(j\omega)^2\vec{E}_s = j\omega\mu(\sigma + j\omega\varepsilon)\vec{E}_s$$

$$\vec{\nabla}^2\vec{E}_s = j\omega\mu(\sigma + j\omega\varepsilon)\vec{E}_s$$

$$\vec{\nabla}^2\vec{H}_s = j\omega\mu(\sigma + j\omega\varepsilon)\vec{H}_s$$

$$\gamma^2 = j\omega\mu(\sigma + j\omega\varepsilon)$$

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} = \alpha + j\beta$$

$$\vec{\nabla}^2\vec{E}_s - \gamma^2\vec{E}_s = 0$$

$$\vec{\nabla}^2\vec{H}_s - \gamma^2\vec{H}_s = 0$$

Helmholtz wave equation
General Media

Propagation Coefficient
Complex quantity

The Plane-wave Solution in Lossy Media

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\vec{a}_x + E_y(x,y,z,t)\vec{a}_y + E_z(x,y,z,t)\vec{a}_z$$

Consider a one-dimensional case (z) with one component only.

$$\frac{\partial^2 E_x}{\partial z^2} - \gamma^2 E_x = 0$$

$$E_x(z) = A\, e^{-\gamma z} + B\, e^{\gamma z}$$

Wave travelling along the $+z$ direction

Wave travelling along the $-z$ direction

Take the $+z$ wave

$$E_x(z) = E_o\, e^{-(\alpha + j\beta)z}$$

$$E_x(z) = E_o\, e^{-\alpha z}\; e^{-j\beta z}$$

Attenuation Phase

Take the sine wave solution

$$E_x(z,t) = E_o \sin(\omega t - \beta z)\, e^{-\alpha z}$$

$$\vec{E}(z,t) = E_o \sin(\omega t - \beta z)\, e^{-\alpha z}\; \vec{a}_x$$

$$\vec{H}(z,t) = H_o \sin(\omega t - \beta z)\, e^{-\alpha z}\; \vec{a}_H$$

The Propagation Coefficient

Starting with the basic definition,

$$\gamma^2 = j\omega\mu(\sigma + j\omega\varepsilon) = -\omega^2\mu\varepsilon + j\omega\mu\sigma$$

complex quantity

$$\gamma = \alpha + j\beta$$

$$\gamma^2 = \alpha^2 + 2j\alpha\beta + \beta^2 = (\alpha^2+\beta^2) + j(2\alpha\beta)$$

Solve for α and β,

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right]}$$

Attenuation Coefficient Np/m

$$\beta = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}+1\right]}$$

Phase Coefficient Rad/m

The Plane-wave Solution in Lossy Media

What about $\vec{H}$?

The EM fields are related using Maxwell's equations.

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \quad \rightarrow \quad \vec{H} = \frac{1}{-j\omega\mu} \vec{\nabla} \times \vec{E}$$

$$\vec{E}(z) = E_o \, e^{-\gamma z} \, \vec{a}_x$$

$$\vec{H}(z) = \frac{1}{-j\omega\mu} \frac{d(E_o \, e^{-\gamma z})}{dz} \vec{a}_y$$

$$\vec{H}(z) = \frac{\gamma}{j\omega\mu} E_o \, e^{-\gamma z} \, \vec{a}_y$$

$$\vec{H}(z) = H_o \, e^{-\gamma z} \, \vec{a}_y$$

$$\eta = \frac{E_o}{H_o} = \frac{j\omega\mu}{\gamma} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

Wave impedance in a lossy medium

$$\eta = |\eta| \angle \theta_\eta$$

$$\tan(2\theta_\eta) = \frac{\sigma}{\omega\varepsilon}$$

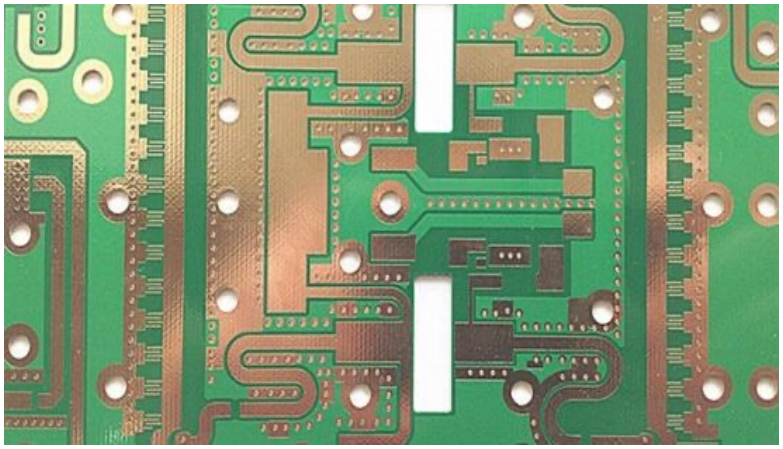
Wave Impedance

For general materials

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = |\eta| \angle\theta_\eta$$

$$|\eta| = \frac{\sqrt{\mu/\varepsilon}}{\left[1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2\right]^{1/4}}$$

$$\tan 2\theta_\eta = \frac{\sigma}{\omega\varepsilon}$$



$$H_o = \frac{E_o}{|\eta| \angle\theta_\eta}$$

A phase may be introduced between E and H due to medium

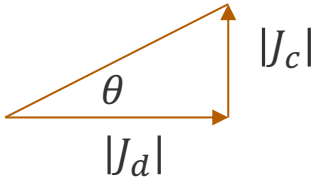
The loss tangent

A measure of how a material is conductive is given by the loss tangent. It is defined as,

$$\tan\theta = \frac{|J_c|}{|J_d|} = \frac{|\sigma E|}{|j\omega\varepsilon E|} = \frac{\sigma}{\omega\varepsilon} = \frac{\varepsilon_r''}{\varepsilon_r'}$$

Loss tangent

$$\theta = 2\theta_\eta$$



Quartz has dielectric constant 3.8, and loss tangent of around 10^{-4} for frequencies from 100 MHz to 10 GHz.

FR4 has dielectric constant 4.4, and loss tangent of 0.030 at 1 GHz.

Classification of Media

Free space $\varepsilon = \varepsilon_o = 8.85\times10^{-12}$ F/m $\mu = \mu_o = 4\pi\times10^{-7}$ H/m $\sigma = 0$

$\eta = \eta_o = 377\ \Omega$ $v = c = 3\times10^{8}$ m/s $\beta = \beta_o = \omega\sqrt{\varepsilon_o\mu_o}$ $\lambda = \lambda_o = 2\pi/\beta_o$

Lossless Medium $\varepsilon = \varepsilon_o\varepsilon_r$ $\mu = \mu_r\,\mu_o$ $\sigma = 0$

$\eta = \eta_o\sqrt{\mu_r/\varepsilon_r}$ $v = c/\sqrt{\varepsilon_r}$ $\beta = \omega\sqrt{\varepsilon_o\varepsilon_r\mu_o}$ $\lambda = 2\pi/\beta$

Lossy/general Medium $\varepsilon = \varepsilon_o\varepsilon_r$ $\mu = \mu_r\,\mu_o$ $\sigma \neq 0$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} \qquad \alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right]} \qquad \beta = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1\right]}$$

$\gamma = \alpha + j\beta$ $v = \omega/\beta$ $\lambda = 2\pi/\beta$

To classify material, use loss tangent

$\frac{\sigma}{\omega\varepsilon} \ll 1$	$\frac{\sigma}{\omega\varepsilon} \gg 1$	$\frac{\sigma}{\omega\varepsilon} \approx 1$
Lossless	Conductors	Lossy material

Classification of Media

Very good conductors $\varepsilon = \varepsilon_o$ $\mu = \mu_r \mu_o$ $\sigma = \infty$

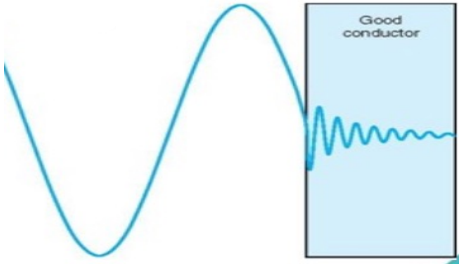
$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \sqrt{\frac{j\omega\mu}{\sigma}} \qquad \alpha = \sqrt{\frac{\omega\mu\sigma}{2}} \qquad \beta = \sqrt{\frac{\omega\mu\sigma}{2}} \qquad v = \sqrt{\frac{2\omega}{\mu\sigma}}$$

Skin depth A measure of how good is a conductor.

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}}$$ Skin depth

$$\eta = \frac{\sqrt{2}}{\sigma\delta} e^{j\pi/4}$$

$$\eta = \frac{1+j}{\sigma\delta}$$



Copper $\sigma = 5.8\times10^7$ S/m

f	10^2	10^8	10^{10}
$\delta\ (mm)$	6.6	$6.6\text{x}10^{-3}$	$6.6\text{x}10^{-4}$

Classification of Dielectric Materials

Dielectrics can be classified based on the nature of the dielectric constant, as follows.

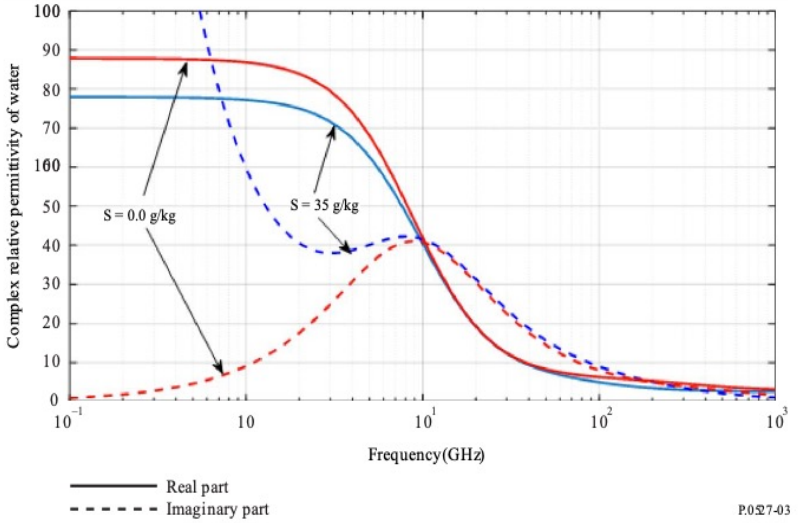
(1) Homogeneous: The value of dielectric constant is the same all over the material; it is independent of position within the material. (otherwise, it is non-homogeneous)

(2) Isotropic: The value of dielectric constant is same along all directions inside the material; it is independent of direction. (otherwise, it is non-isotropic)

(3) lossy: The value of dielectric constant is a complex number. (otherwise, it is lossless, and the dielectric constant is a real number)

(4) Dispersive: The value of dielectric constant is dependent on frequency. (Example, sea water, PDMS, FR4).

(5) Linear: The value of dielectric constant is independent of magnitude of applied electric field. (otherwise, it is non-linear)



Complex ε_r

We can combine the conductivity σ in the definition of the dielectric constant ε_r . In this case, a complex dielectric constant.

The time-harmonic Maxwell's equation is $$\vec{\nabla} \times \vec{H} = \sigma \vec{E} + j\omega\varepsilon \vec{E}$$

Combine terms $$\vec{\nabla} \times \vec{H} = (\sigma + j\omega\varepsilon)\vec{E} = j\omega\,\varepsilon\left(1 - \frac{j\sigma}{\omega\varepsilon}\right)\vec{E}$$

ε_{rc}

Compare to the lossless equation $$\varepsilon_{rc} = \varepsilon\left(1 - \frac{j\sigma}{\omega\varepsilon}\right) = \varepsilon_o\varepsilon_r\left(1 - \frac{j\sigma}{\omega\varepsilon_o\varepsilon_r}\right)$$

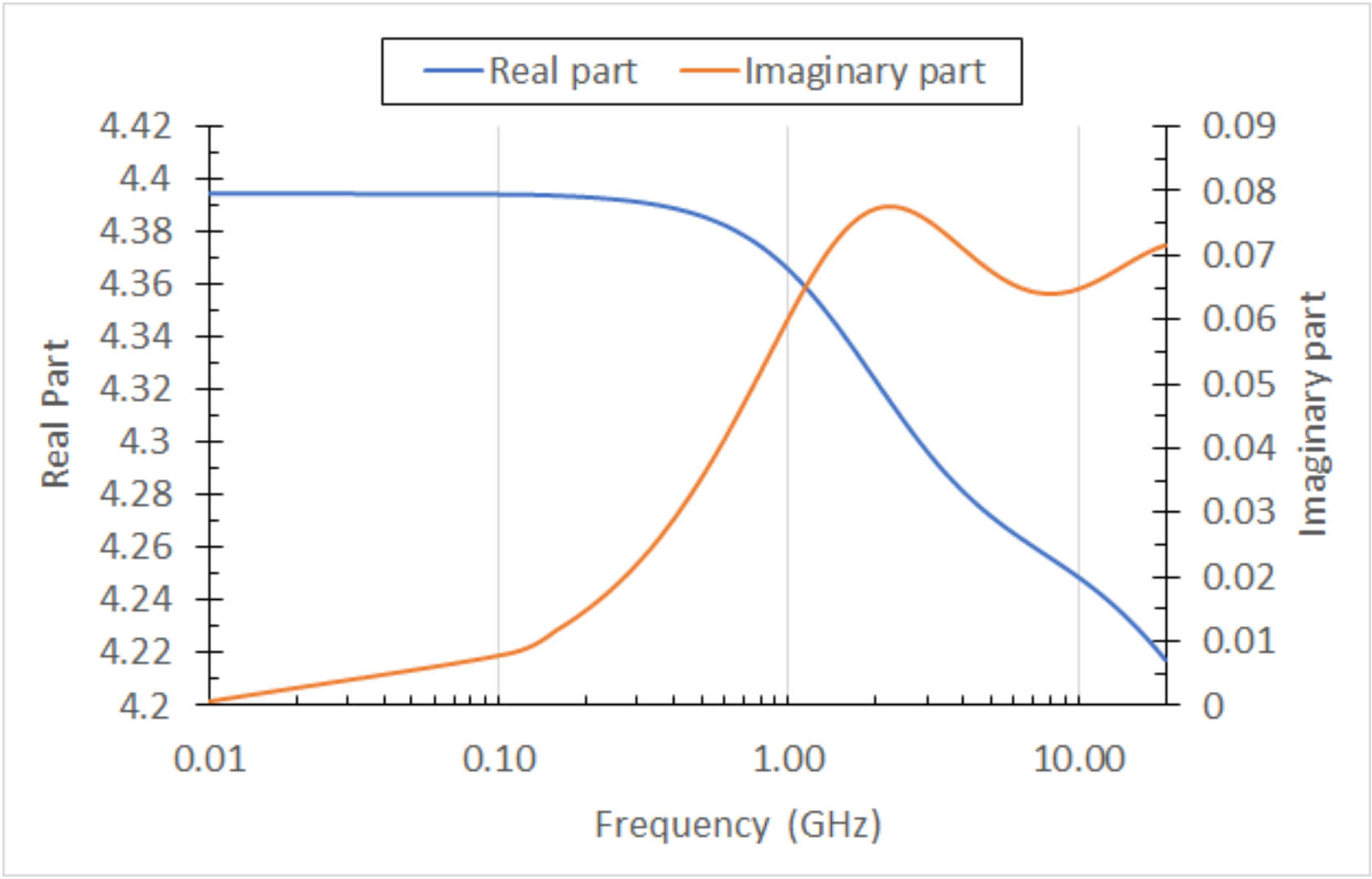
$$\varepsilon_c = \varepsilon_o\left(\varepsilon_r - j\frac{\sigma}{\omega\varepsilon_o}\right)$$

ε_{rc}

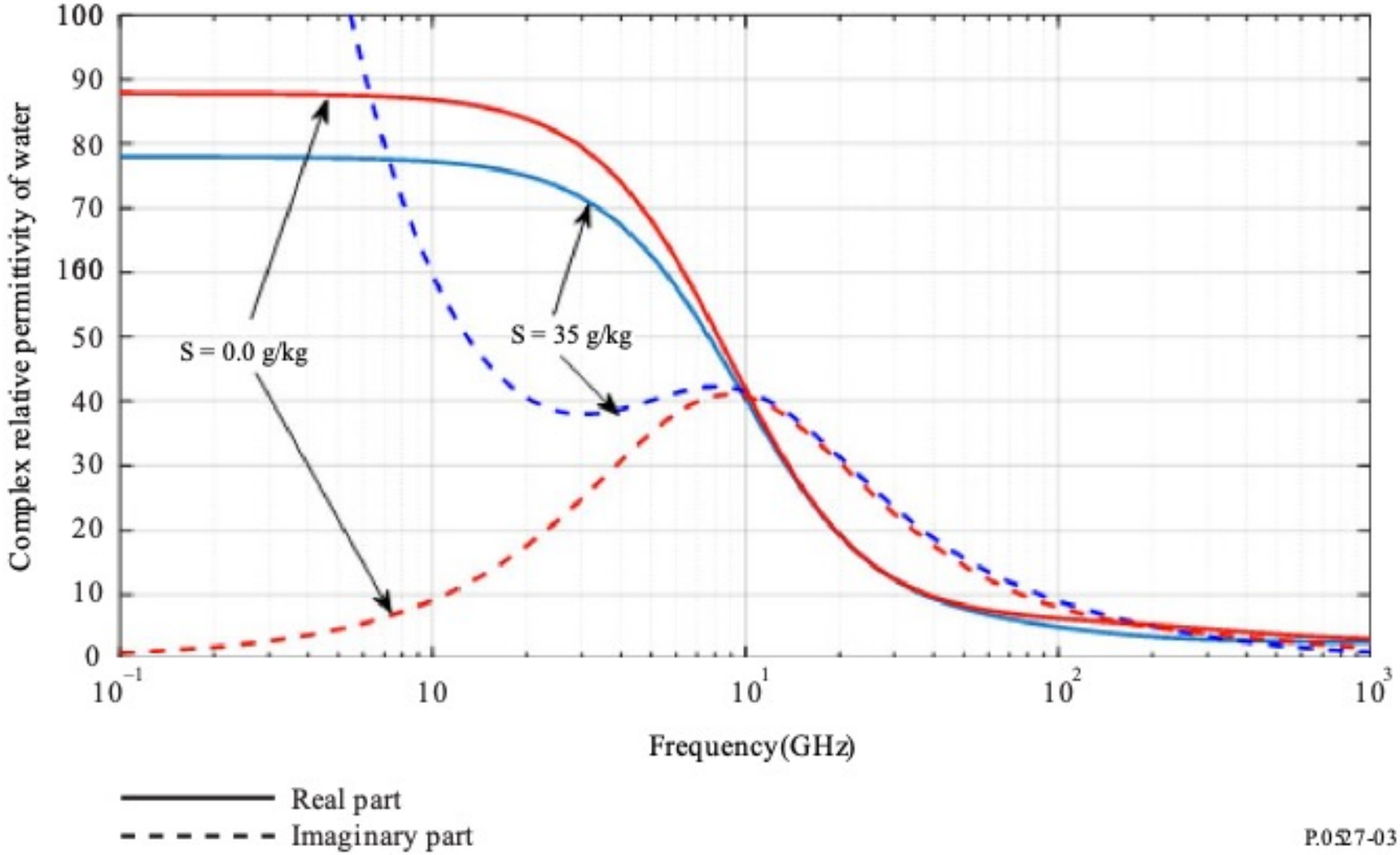
$$\varepsilon_{rc} = \varepsilon_r' - j\varepsilon_r''$$

Complex dielectric constant

FR4 Dielectric Function



Sea Water Dielectric Function



Examples

(1) Consider an EM wave travelling in a lossy dielectric of impedance $200\angle 30°\ \Omega$ with some frequency. At this frequency, the magnetic field of the wave is given by

$$\vec{H}(x,t) = 10\, e^{-\alpha x} \cos(\omega t - x/2)\ \vec{a}_y \quad \text{(A/m)}$$

The direction of propagation is $+x$ direction $\vec{a}_k = \vec{a}_x$.

Expression for E. $\quad E_o = \eta H_o \quad E_o = 200\angle 30°\ (10) = 2000\, e^{j\pi/6}$ V/m

$\beta = 1/2$ rad/m $\quad \vec{a}_k = \vec{a}_E \times \vec{a}_H \quad \vec{a}_x = \vec{a}_E \times \vec{a}_y \rightarrow \vec{a}_E = -\vec{a}_z$

$$\frac{\alpha}{\beta} = \left(\frac{\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1}{\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1} \right)^{1/2} = \frac{1}{\sqrt{3}}$$

Note that $\tan 2\theta_\eta = \frac{\sigma}{\omega\varepsilon} = \sqrt{3}$

$\alpha = \frac{1}{2\sqrt{3}}$ Np/m $\quad \vec{E}(x,t) = 2\, e^{j\pi/6}\, e^{-\frac{x}{2\sqrt{3}}} \cos(\omega t - x/2)\ (-\vec{a}_z)$ kV/m

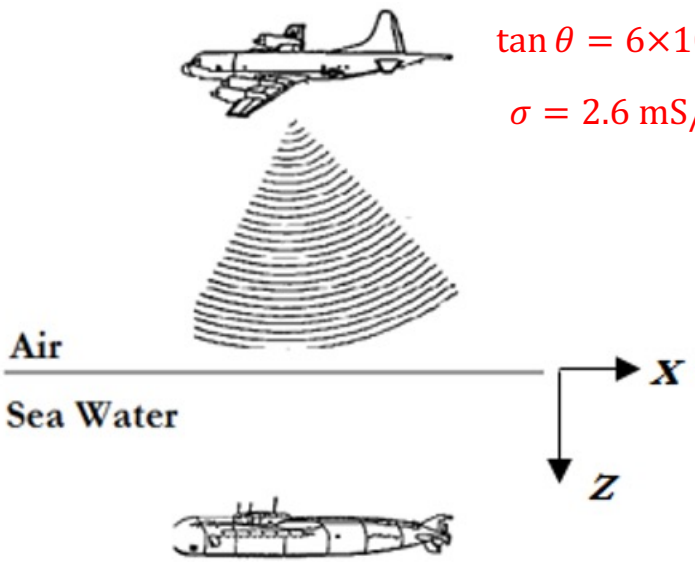
Examples

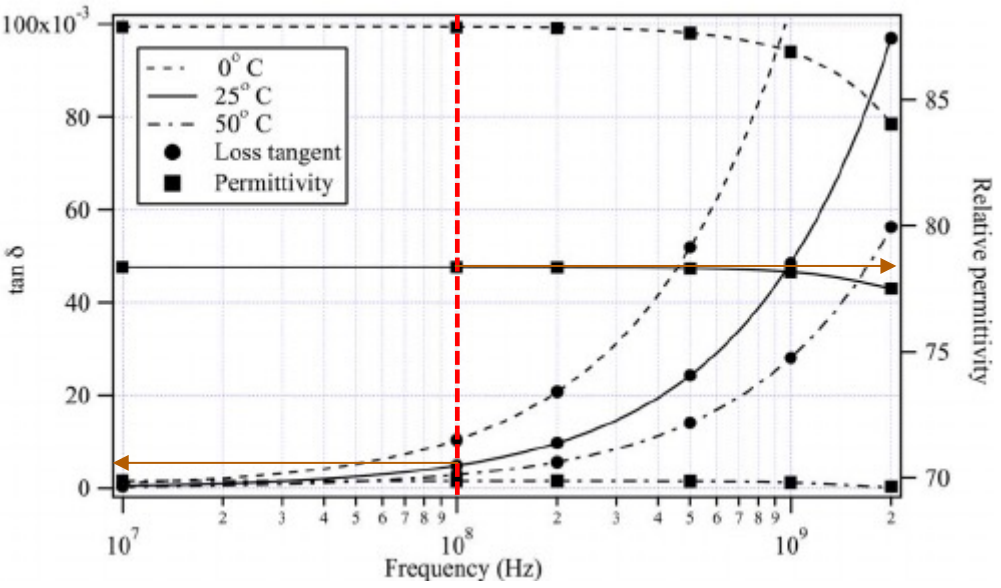
(2) A 100-MHz antenna carried by an airplane flying over the ocean surface generates a wave that approaches the flat water surface in the form of a plane wave. At this frequency, seawater is characterized by. At the water side of the interface, the electric field is x-polarized with amplitude of 3 kV/m. Just below the ocean surface, a submarine tries to read the message sent by the airplane. The submarine receiver requires at least an amplitude of 0.01 μV/m for a valid reception. At what maximum depth below ocean surface should the submarine be for successful communication with the airplane?

$\varepsilon_r = 78$

$\tan\theta = 6\times10^{-3}$

$\sigma = 2.6$ mS/m





Q & A

Send me an email if you have any questions.